

of Questions: 30

Question #: 1

A 67-year-old man comes to the physician because he is unable to close his right eye. Physical examination shows left eyelid is able to close without any problem. Which of the following nerve lesions most likely will lead to inability to close the eye?

- A. CN III
- B. CN VII
- C. CN V3
- D. CN II
- E. Cervical Sympathetic chain

Item Description: CN III nerve lesion: ptosis

Question #: 2

A 43-year-old patient comes to the physician because the tongue deviates toward the right side, after endarterectomy procedure. Physical examination shows the sensory innervation of the head and neck soft tissues, as well as the laryngeal muscles, appears unaffected. Which of the following structures is most likely damaged by endarterectomy procedure?

- A. Hypoglossal nerve and hypoglossal canal
- B. Mandibular nerve and foramen ovale
- C. Facial nerve and internal acoustic meatus
- D. Glossopharyngeal nerve and jugular foramen
- E. Facial nerve and stylomastoid foramen

Item Description: Hypoglossal nerve and hypoglossal canal

Question #: 3

A 24-year-old man come to physicians office because he collapsed to the ground after a ball hit right side of his head and was unconcious for about four minutes. Once concious he expericend with a severe headache and dousy. Physical examination shows bruise on right temple and dilated right pupil. Radiological examination shows a fracture of the squamous part of the right temporal bone. and an epidural hematoma. Which of the following structures is most likely damaged?

- A. Middle meningeal artery
- B. Accessory meningeal artery
- C. Infraorbital artery
- D. Sphenopalatine artery

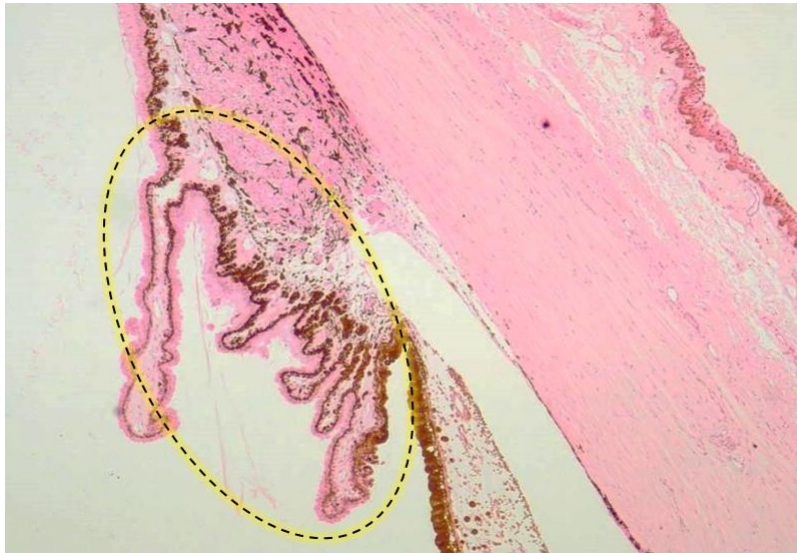
Item Description: Middle meningeal artery

Question #: 4

A 42-year-old woman comes to the physician because of a 2-day history of blurred vision. Her temperature is 100.4°F (38°C). Physical examination shows no structural abnormalities. The physician suspects uveitis (inflammation of the uveal tract). A normal photomicrograph of the anterior portion of the affected structure is shown. Which of the following is a characteristic feature of the circumscribed structure?

- A. Has epithelial cells that produce and secretes aqueous humor
- B. Contains smooth muscles that reduces the size of the pupil
- C. Contains smooth muscles that increases the size of the pupil
- D. Possesses the Meibomian glands
- E. Synthesizes and secretes the vitreous humor

Attachment:



attachment_for_itemid_342603.jpg

Item Description: HCB.1338 - Ciliary body

Question #: 5

A 24-year-old woman comes to the physician because of a 1-hour history of severe upper abdominal pain. She has a 2-year-history of recurrent heartburn. Endoscopy is done and a biopsy is obtained from the lower end of esophagus. Which of the following epithelia most likely normally lines this region?

- A. Simple cuboidal
- B. Simple squamous
- C. Pseudostratified
- D. Stratified squamous keratinized
- E. Stratified squamous non-keratinized

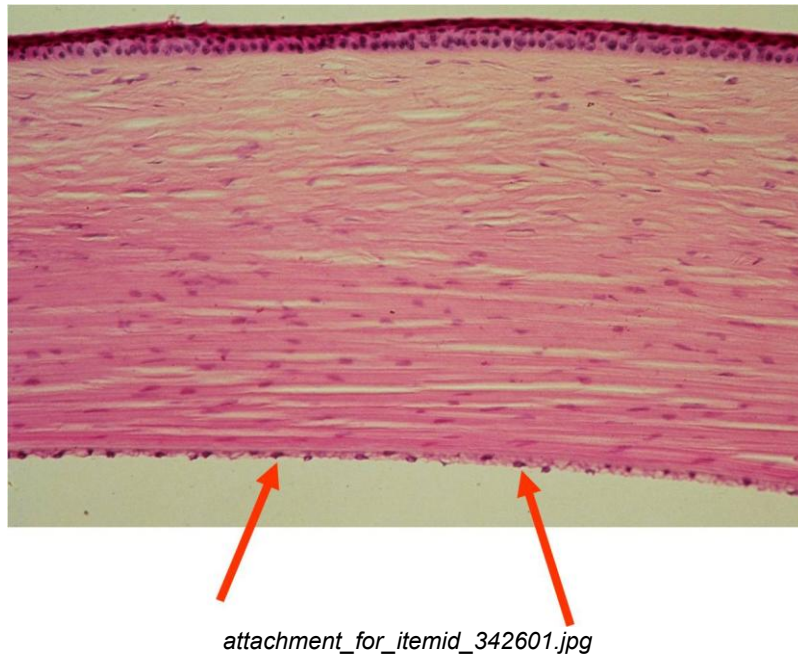
Item Description: HCB.1206 - Esophagus: Epithelium

Question #: 6

A 70-year-old man is brought to the physician because of a 4-month history of a progressive loss of vision in both eyes. Physical examination shows reduced visual acuity in both eyes. A slit lamp examination of both eyes shows inclusion bodies within the cells shown at the tip of the arrows in the attached photomicrograph. Which of the following best describes a function of these cells?

- A. Maintenance of the shape of the eyes
- B. Prevent posterior spread of infection
- C. Maintain optimal hydration of the corneal stroma
- D. Absorption of stray light
- E. Adjustment of pupil diameter
- F. Maintenance of the shape of the lens

Attachment:



Item Description: HCB.1334 - Corneal endothelium

Question #: 7

A 45-year-old man comes to the physician because of a 3-week history of yellowish discoloration of the eyes. Physical examination shows jaundice and ascites. Laboratory studies are shown. Which of the following best explains the pathogenesis of this disorder?

- A. Increased neutrophil elastase activity
- B. Decreased dietary copper absorption
- C. Increased dietary iron absorption
- D. Decreased biliary copper excretion
- E. Mutation in membrane-bound gated-chloride ion channel
- F. Decreased iron availability for erythropoiesis

Attachment:

	Patient	Normal range
Alanine aminotransferase (ALT)	180	40-60
Aspartate aminotransferase (AST)	250	40-60
Alkaline phosphatase (ALP)	140	50-70
Gamma glutamyl transferase (GGT)	130	50-70
Serum ferritin	340	20-250 ng/mL

attachment_for_itemid_218394.jpg

Item Description: Hemochromatosis

Question #: 8

A 46-year-old woman comes to the physician because of a 3-month history of yellowish discoloration of the eyes and skin. Laboratory and imaging studies show obstruction of the common bile duct by gallstones. Which of the following laboratory tests best reflects this patient's condition?

- A. I
- B. II
- C. III
- D. IV
- E. V

Attachment:

	Type of serum bilirubin	Urine bilirubin	Urine urobilinogen
I	Indirect and direct	Present	Decreased
II	Unconjugated and conjugated	Absent	Increased
III	Unconjugated	Absent	Increased
IV	Direct	Absent	Absent
V	Conjugated	Present	Absent

attachment_for_itemid_140930.jpg

Item Description: Posthepatic jaundice

Question #: 9

A 9-month-old girl is brought to the emergency room because of a 2-hour history of drowsiness and refusal of feeds. Two days earlier, she had fever, vomiting and diarrhea. Urinalysis shows octanoyl (C-8) and decanoyl (C-10) carnitine derivatives. Which of the following is the most likely diagnosis?

- A. von Gierke disease
- B. McArdle syndrome
- C. Myopathic carnitine deficiency
- D. Medium chain acyl CoA dehydrogenase deficiency
- E. Defective peroxisomal biogenesis
- F. Maple syrup urine disease

Attachment:

ANALYTE	Patient	Reference Range
Glucose (mg/dL)	40	70 - 110
Acetoacetate (mg/dL)	0.1	0.2 - 1.0

attachment_for_itemid_140795.jpg

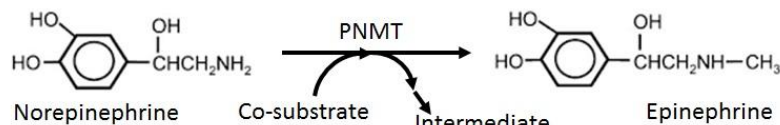
Item Description: MCADD

Question #: 10

An investigator is studying catecholamine neurotransmitters and focuses on the formation of epinephrine from norepinephrine. It is found that an intermediate labeled in the figure is formed that must be removed. If metabolism of this intermediate is blocked, a compound will accumulate which will result in associated pathologic changes. A figure depicting the catalyzed reaction is shown. Which of the following intermediate molecules is most likely formed when this reaction takes place?

- A. Carbon monoxide
- B. Phenylpyruvic acid
- C. Homocysteine
- D. Nitric oxide
- E. Methylmalonic acid
- F. Delta-aminolevulinate
- G. Citrulline

Attachment:



attachment_for_itemid_112187.JPG

Item Description: QUIZ - formation of catecholamines

Question #: 11

A 61-year-old man comes to the physician because of a 1-week history of shortness of breath and productive cough. He has smoked 1 pack of cigarettes daily for 40 years. Physical examination shows diffuse rhonchi and wheezing. A chest x-ray and pulmonary function tests confirm the diagnosis of an acute exacerbation of chronic obstructive lung disease. He is administered a drug that stimulates adrenergic receptors in the bronchial smooth muscles producing bronchodilation. Which of the following intracellular mediators is most likely increased in the bronchial smooth muscles of this patient?

- A. cAMP
- B. cGMP
- C. DAG
- D. Protein kinase C
- E. Nitric oxide

Item Description: PHYS.0044 Adrenergic receptors Cellular signaling

Question #: 12

A 29-year-old man comes to the physician because of a 3-month history of headaches and decreased libido. His vital signs are within normal limits. Physical examination shows bitemporal hemianopsia, bilateral gynecomastia, and scanty hairs on face, groin and axilla. Laboratory studies show reduced serum levels of testosterone. MRI of the brain shows enlargement of the Sella turcica. Which of the following hormones is most likely inhibited in this patient?

- A. Thyroid stimulating hormone
- B. Growth hormone
- C. Prolactin
- D. Gonadotropin releasing hormone
- E. Oxytocin

Item Description: PHYS.1017. Prolactinoma laboratory findings

Question #: 13

A 43-year-old man comes to the physician because of a 3-month history of recurrent episodes of headaches, blurry vision, and exertional dyspnea. His blood pressure is 156/96 mm Hg; other vital signs are within normal limits. Physical examination shows a loud S2 in aortic area. Laboratory studies show

elevated serum sodium, reduced serum potassium, and elevated serum bicarbonate. CT scan of the abdomen shows a mass in the right adrenal gland. He is diagnosed with an aldosterone-secreting tumor. Increased activity of which of the following renal tubular transport mechanisms is most likely in this patient?

- A. Sodium-Glucose secondary active transport in the proximal convoluted tubules
- B. Sodium-Potassium-Chloride transporter in the thick ascending limb of loop of Henle
- C. Sodium-Amino acid transporter in the proximal convoluted tubules
- D. Epithelial Sodium channels in the collecting duct
- E. Sodium-chloride transporter in the distal convoluted tubules

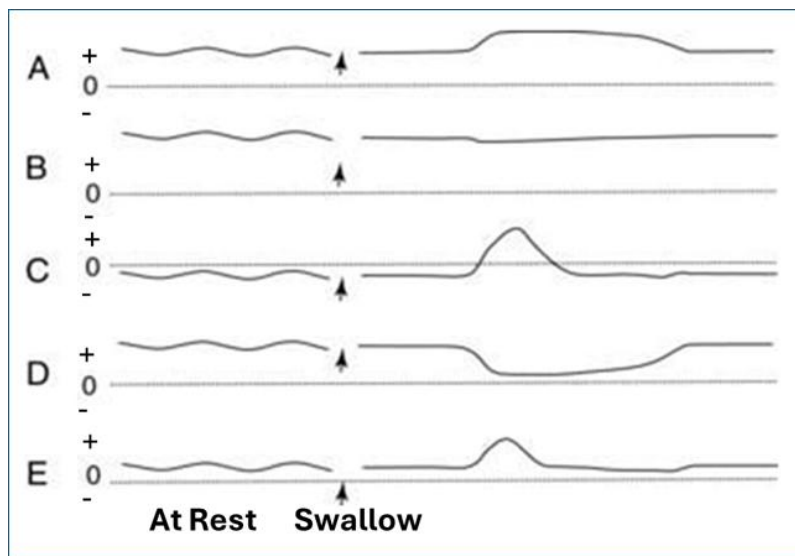
Item Description: PHYS.1028.Aldosterone action

Question #: 14

A 58-year-old man comes to the physician because of a 3-week history of difficulty swallowing solids. He does not have any chronic illnesses. Physical examination shows no abnormalities. An upper GI endoscopy shows no abnormalities. An esophageal manometry shows normal findings. Mid-esophageal manometry findings in normal and diseased conditions are shown in the attached figure. Which of the following manometric recordings best represents this patient?

- A. A
- B. B
- C. C
- D. D
- E. E

Attachment:



attachment_for_itemid_141163.png

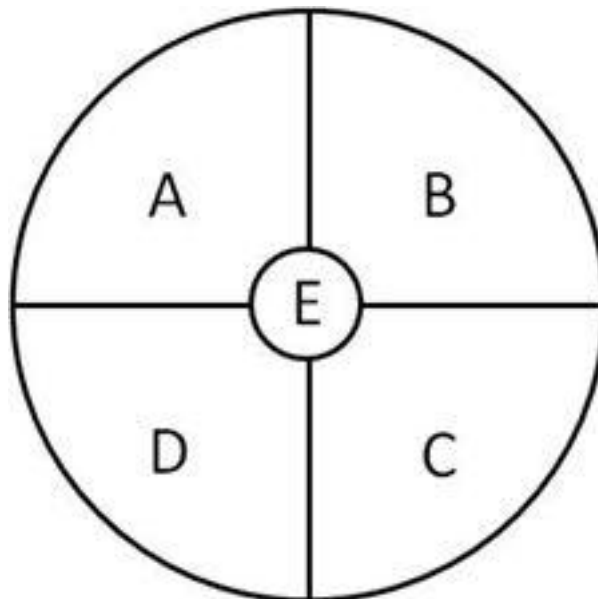
Item Description: PHYS.1101. Esophageal Motility

Question #: 15

An 82-year-old woman comes to the physician because of a 9-month history of visual problems. Fundoscopic examination shows retinal mottling consistent with age-related macular degeneration. Which of the following portions of the visual field is most likely disrupted in the patient (see attached figure)?

- A. A
- B. B
- C. C
- D. D
- E. E

Attachment:



attachment_for_itemid_345346.jpg

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0222 Visual field retina

Question #: 16

A 24-year-old man comes to the physician because of a 3-month history of double-vision and difficulty seeing with his left eye. His left pupil is dilated, and his left eyelid droops. His left eye is tonically deviated laterally and down. Residual function of which of the following extraocular muscles most likely mediates the downward component of the deviation of the eye?

- A. Superior rectus
- B. Inferior rectus
- C. Superior oblique

- D. Inferior oblique
- E. Lateral rectus
- F. Medial rectus

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0230 Extraocular function

Question #: 17

A 49-year-old man comes to the physician because of a 1-day history of hearing loss. Weber's test lateralizes to the left. Rinne's test shows AC > BC bilaterally. Which of the following structures is most likely lesioned?

- A. Right cingulate
- B. Right ossicle chain
- C. Right cochlea
- D. Right caudate
- E. Right insula
- F. Left cingulate
- G. Left ossicle chain
- H. Left cochlea
- I. Left caudate
- J. Left insula

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0239 Auditory pathways

Question #: 18

A 27-year-old man comes to the physician because of a 2-week history of hearing problems. He says that a fire-cracker had exploded near his left ear. Tuning fork tests show sensorineural hearing loss affecting the left ear and a diagnosis of acquired hearing loss is made. Which of the following findings during Weber's test is most likely?

- A. Lateralization to the left ear
- B. Lateralization to the right ear
- C. Bilaterally normal hearing
- D. BC = AC = 0 bilaterally
- E. BC > AC for the right ear

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0239 Auditory pathways

Question #: 19

A 47-year-old man comes to the physician because of a 5-day history of headaches. Assessment of pupillary light reflexes shows a loss of pupillary constriction bilaterally with presentation of light to his right eye. Both pupils constrict with the presentation of light to his left eye. Which of the following structures is most likely lesioned in this patient?

- A. Left CN III
- B. Right CN III
- C. Left CN II
- D. Right CN II
- E. Left CN V
- F. Right CN V

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0210 Pupillary light reflex

Question #: 20

A 57-year-old man is brought to the emergency department because of a 2-hour history of problems seeing. Confrontation testing shows a visual field defect. Imaging studies show an infarction involving the occipital branch of the right posterior cerebral artery. Which of the following visual field defects is most likely in this patient?

- A. Bitemporal hemianopia
- B. Left superior homonymous quadrantanopia
- C. Right superior homonymous quadrantanopia
- D. Left inferior homonymous quadrantanopia
- E. Right inferior homonymous quadrantanopia
- F. Bitemporal hemianopia with macular sparing
- G. Left homonymous hemianopia
- H. Right homonymous hemianopia
- I. Left homonymous hemianopia with macular sparing
- J. Right homonymous hemianopia with macular sparing

Item Description: SOM.MK.I.BPM2.1.NB.4.NSCI.0228 Visual pathway lesions

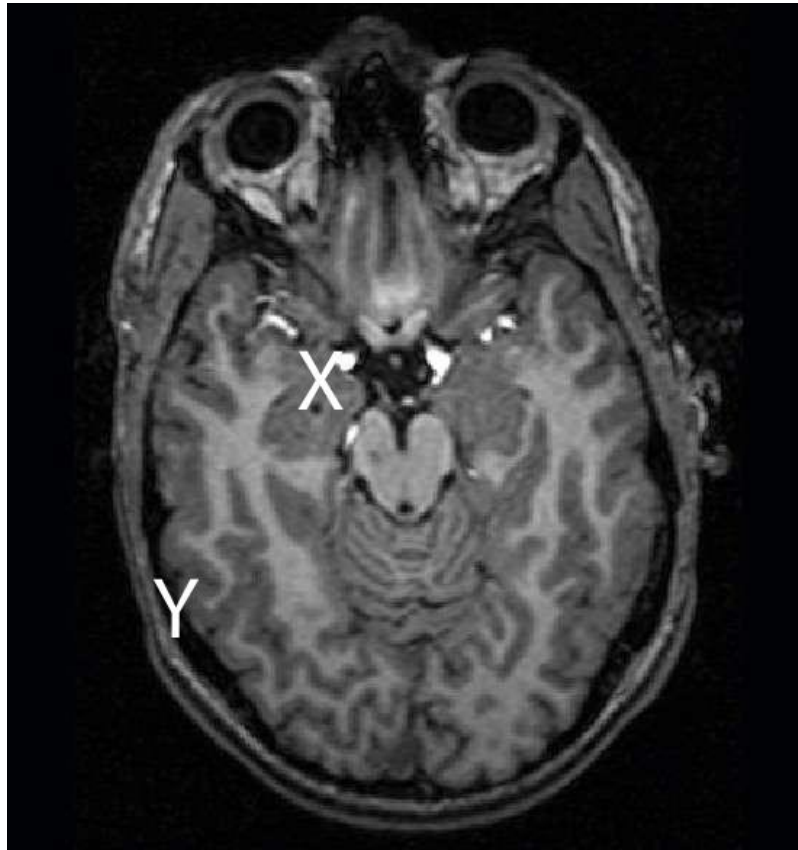
Question #: 21

A 52-year-old man is brought to the emergency department because of a 2-hour history of a traumatic head wound. Imaging studies show that a fast-moving fragment of metal entered the skull at Site Y and

lodged at Site X. Which of the following trajectories is most likely followed by the fragment as it passes through the brain?

- A. Anteromedial
- B. Posteromedial
- C. Anterolateral
- D. Posterolateral
- E. Dorsomedial
- F. Ventromedial
- G. Dorsolateral
- H. Ventrolateral

Attachment:



attachment_for_itemid_345340.jpg

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0009 Neuroimaging anatomical vectors

Question #: 22

A 47-year old man is brought to the emergency department because of a 3-hour history of nausea and a fainting spell. Imaging studies show hypoperfusion in a small localized area of the brain. An infarction is suspected. Which of the following cells will most likely proliferate to form a scar in the injured area?

- A. Schwann cells
- B. Microglia
- C. Oligodendrocytes
- D. Astrocytes
- E. Ependymocytes

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0017 Glial cells

Question #: 23

A 49-year-old man comes to the physician because of a 1-day history of hearing loss. Weber's test lateralizes to the left. Rinne's test shows AC > BC bilaterally. Which of the following structures is most likely lesioned?

- A. Right cingulate
- B. Right ossicle chain
- C. Right CN VIII
- D. Right caudate
- E. Right insula
- F. Left cingulate
- G. Left ossicle chain
- H. Left CN VIII
- I. Left caudate
- J. Left insula

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0239 Auditory pathways

Question #: 24

A 44-year-old man comes to the physician because of an 8-hour history of clumsiness. Imaging studies show a lesion overlapping with the left cerebocerebellum. Which of the following clinical manifestations is most likely in this patient?

- A. Left-sided tingling
- B. Right-sided tingling
- C. Left dysdiadochokinesia
- D. Right dysdiadochokinesia
- E. Left resting tremor
- F. Right resting tremor

Item Description: SOM.MK.I.BPM2.4.NB.2.NSCI.0202 Cerebellar lesions and manifestations

Question #: 25

A 68-year-old woman comes to the physician because of a 6-month history of blurry vision. Fundoscopic studies show bilateral macular lesions. Which of the following processes is most likely?

- A. Loss of lenticular elasticity
- B. Occipital stroke
- C. Neurodegeneration
- D. Demyelination
- E. Diabetic optic neuritis

Item Description: SOM.MK.I.BPM2.4.NB.4.NSCI.0215 Causes of visual impairment

Question #: 26

A 78-year-old man comes to the physician because of a 4-day history of clumsiness of his right arm and leg. He also has difficulty maintaining his balance in the dark. Physical examination shows diminished touch, vibration and conscious proprioception for the right arm and leg. Imaging studies show a spinal infarction. Which of the following arterial regions is most likely implicated?

- A. Anterior spinal artery (cervical level)
- B. Left posterior spinal artery (thoracic level)
- C. Left posterior spinal artery (cervical level)
- D. Right posterior spinal artery (thoracic level)
- E. Right posterior spinal artery (cervical level)

Item Description: SOM.MK.I.BPM2.5.NB.5.NSCI.0364 Spinal lesions and sensory loss

Question #: 27

A 27-year-old man comes to the physician because of a 6-month history of short, unilateral attacks of severe, shooting facial pain, that feel "like an electric shock." Each pain attack lasts only a few seconds, but they can cluster, followed by pain-free intervals lasting for several weeks. The pain attacks can be triggered by light touch near the left corner of the mouth and by eating or talking. MRI of the head is normal. Dental conditions are normal. The pain is refractory to pharmacological intervention. Surgical ablation of pseudounipolar cells is considered. Which of the following structures is most closely associated with the proximal root of the surgically targeted cell bodies?

- A. Cerebrum
- B. Diencephalon
- C. Midbrain
- D. Pons
- E. Medulla

Item Description: SOM.MK.IV.BPM2.4.NB.4.NSCI.0126 Pain management

Question #: 28

A 54-year-old man comes to the physician because of an 8-hour history of clumsiness. Imaging studies show a lesion overlapping with the right lateral hemisphere. Which of the following clinical manifestations is most likely?

- A. Right-sided tingling
- B. Left-sided tingling
- C. Left dysdiadochokinesia
- D. Right dysdiadochokinesia
- E. Left resting tremor
- F. Right resting tremor

Item Description: SOM.MK.I.BPM2.4.NB.2.NSCI.0202 Cerebellar lesions and manifestations

Question #: 29

A 21-year-old man is brought to the emergency department because of a 40-minute history of a traumatic injury following a fragment of metal penetration to his back. CT scans of his trunk show a right spinal hemisection at T10. Which of the following manifestations is most likely?

- A. Loss of nociception of the left leg
- B. Loss of proprioception of the left leg
- C. Weakness of the left leg
- D. Loss of nociception of left dermatome T10
- E. Loss of nociception of the right arm

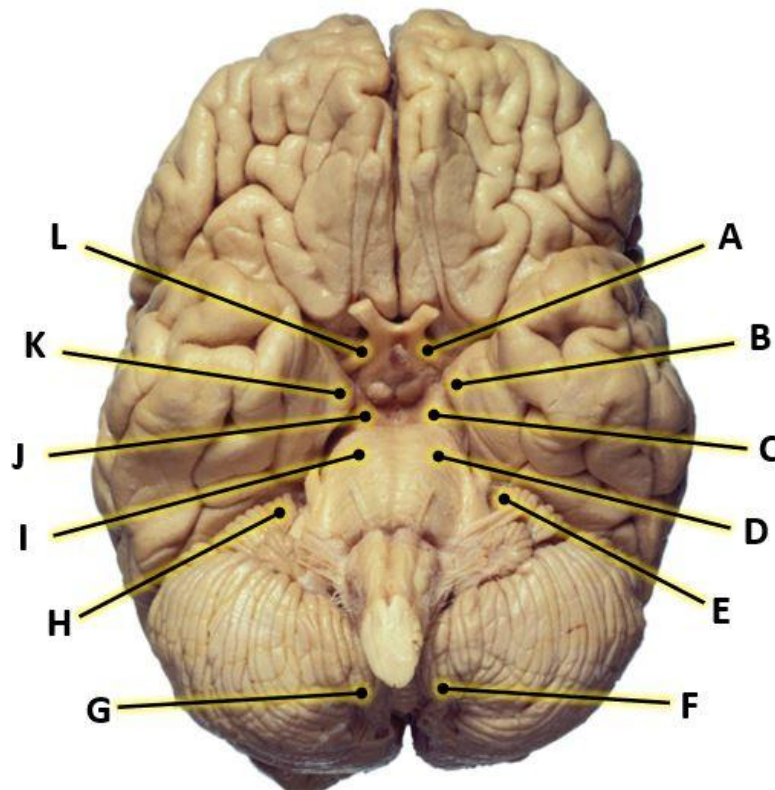
Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0119 Use knowledge of the anatomical and functional organization of the somatosensory system to identify and localize the clinical correlations Shingles (herpes zoster), Brown-Sequard syndrome and Syringomyelia

Question #: 30

A 54-year-old man is brought to the emergency department because of weakness on the right side of his body, including upper and lower extremities that developed over the past few months. Neurologic examination confirms a right hemiplegia. His left eye pupil is abnormally dilated and responds sluggishly to light. Left eye movements are deficient. Herniation of which of the structures marked in the brain image attached through the tentorial notch is the most likely cause of the pathological findings in this patient?

- A. A
- B. B
- C. C
- D. D
- E. E
- F. F
- G. G
- H. H
- I. I
- J. J
- K. K
- L. L

Attachment:



attachment_for_itemid_345331.JPG

Item Description: Location of the uncus, uncal herniation